Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	15 25	**1**	
1(b)	05 50	**1**	
2(a)	94	**1**	
2(b)	Correct circle drawn	**2**	**B1** for centre of *AB* marked
3(a)	−4	**2**	**B1** for −12 seen or **M1** for −7 − 5 + 8 or better
3(b)	T – 2 cao	**1**	
4	129 Angles at a point sum to 360 oe	**2**	**B1** for each
5(a)(i)	$\frac{3}{8}$ oe	**1**	
5(a)(ii)	1	**1**	
5(b)	60	**1**	**FT** *their* **(a)(i)**
6	2 678 400	**2**	**M1** for [31 ×] 24 × 60 × 60 or for 31 × 24 [× 60][× 60] s
7	Correct net	**3**	**B2** for 4 correctly connected faces drawn or **B1** for 2 correct connected faces drawn or for 6 correct faces drawn
8(a)	$\frac{1}{40}$ or 0.025	**1**	
8(b)	3.4200 cao	**2**	**B1** for 3.419 to 3.420
8(c)	$4[.0] \times 10^{[1]}$ cao	**1**	
9(a)	2	**1**	
9(b)	– 7 – 3 3	**2**	**B1** for 2 correct

Question	Answer	Marks	Partial Marks
9(c)	Correct ruled graph	**1**	**FT** *their* **(b)** provided it is a single ruled straight line
10	$\begin{pmatrix} -4 \\ 3 \end{pmatrix}$	**1**	
11(a)	1 | 3 5 8 2 | 1 4 6 8 3 | 6 8 8 4 | 5 8	**2**	**M1** for correct but not ordered or for two correct rows ordered
11(b)	27	**1**	
12	32	**2**	**M1** for $\frac{112}{12-5}$
13	$\frac{15}{28}\times\frac{7}{4}$ or $\frac{15}{28}\div\frac{16}{28}$ oe	**M1**	
	$\frac{15}{16}$ cao	**A2**	**A1** for $\frac{105}{112}$ oe
14(a)	45.9	**2**	**M1** for $0.5 \times 8.5 \times 10.8$ oe
14(b)	33[.0] or 33.04…	**3**	**M2** for $8.5 + 10.8 + \sqrt{8.5^2 + 10.8^2}$ oe or **M1** for $8.5^2 + 10.8^2$ oe
15	35 661 nfww	**3**	**M1** for $30\,000 \times \left(1+\frac{2.5}{100}\right)^7$ oe **A1** for 35 660.57 or correct evaluation but not rounded to the nearest dollar **B1** for correctly rounding *their* answer to the nearest dollar
16(a)	5	**1**	
16(b)	8	**1**	
17	2.77 or 2.768…	**2**	**M1** for $\sin[37 =]\ \frac{h}{4.6}$ or better
18(a)	$3x(x - 4y)$ final answer	**2**	**B1** for $3(x^2 - 4xy)$ or $x(3x - 12y)$
18(b)	$m^2 - m - 6$ final answer	**2**	**M1** for 3 terms from m^2, $-3m$, $+2m$, -6